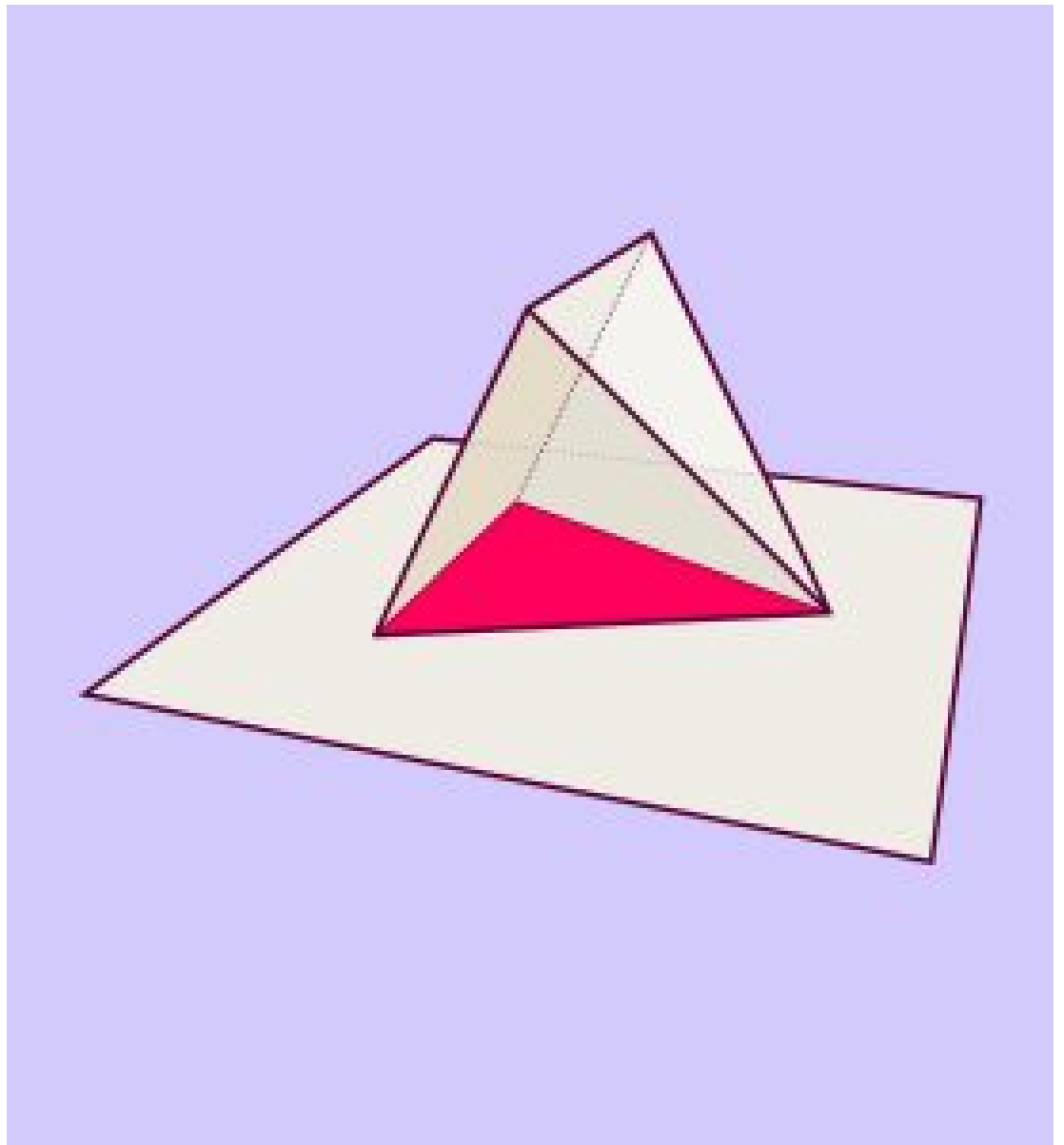


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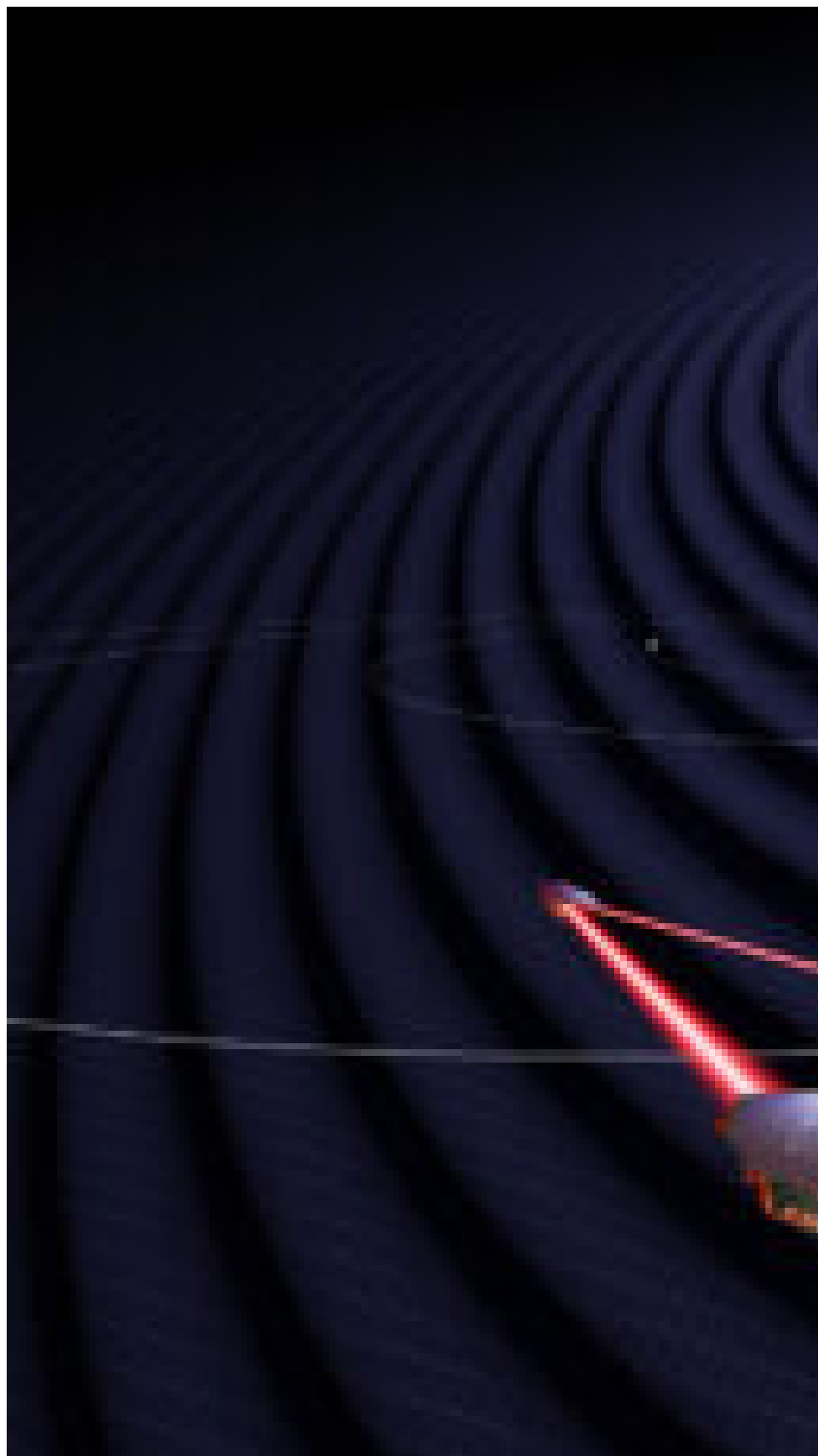
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[Simplicity in higher dimensions](#)

According to theories such as string theory, our world has more than the usual three spatial dimensions. On the one hand, that prediction leads to problems - physicists have to explain how it comes that, in everyday life, we experience only three of the ten space ...

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Observing gravitational waves in space with LISA

Earth-based instruments have already observed many dozens of gravitational-wave signals from merging compact objects such as binary systems of black holes and neutron stars. Their future upgrades and third-generation successors will detect many thousands more such ...

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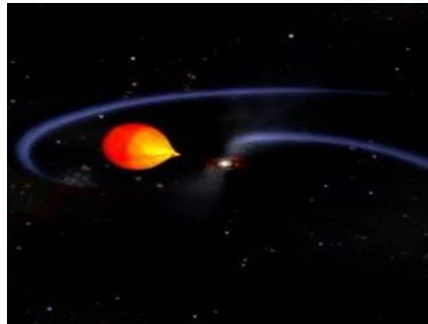




[Third-generation gravitational-wave detectors](#)

The era of gravitational wave astronomy has begun – with Advanced LIGO and Advanced Virgo having observed 90 signals as of June 2022. In the coming years, these kilometer-sized and high-precision observatories, together with the Japanese KAGRA and the planned LIGO ...

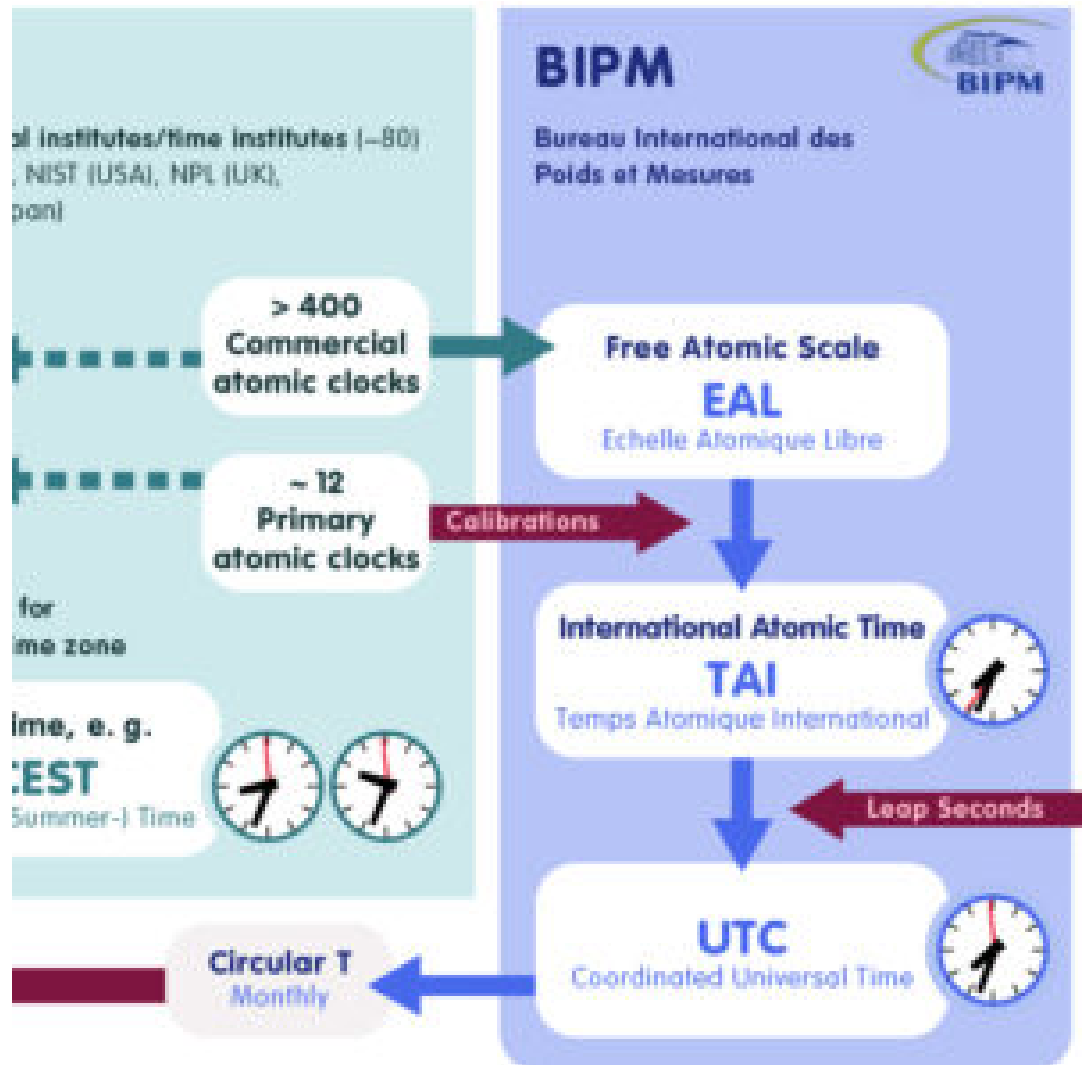
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Continuous gravitational waves

Cosmic lighthouses born in supernova explosions When stars at least eight times as massive as our Sun run out of nuclear fuel, their lives end in supernovae. These explosions bring forth new cosmic objects: neutron stars. Because of their strong magnetic fields and ...

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How time is made

In science and technology, time is – as Einstein knew – relative: there is no generally valid definition of simultaneity in nature that can be derived solely from the laws of physics. And yet, in practice, it is very important to agree on a uniform time, at least ...

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[Relativity and satellite navigation](#)

At sea or in the wilderness, precise positioning can be vital. But even in everyday life, there is not always a place or street sign nearby when the question comes up: Where are we now? The answer is readily available these days. A small device, not much bigger than a ...

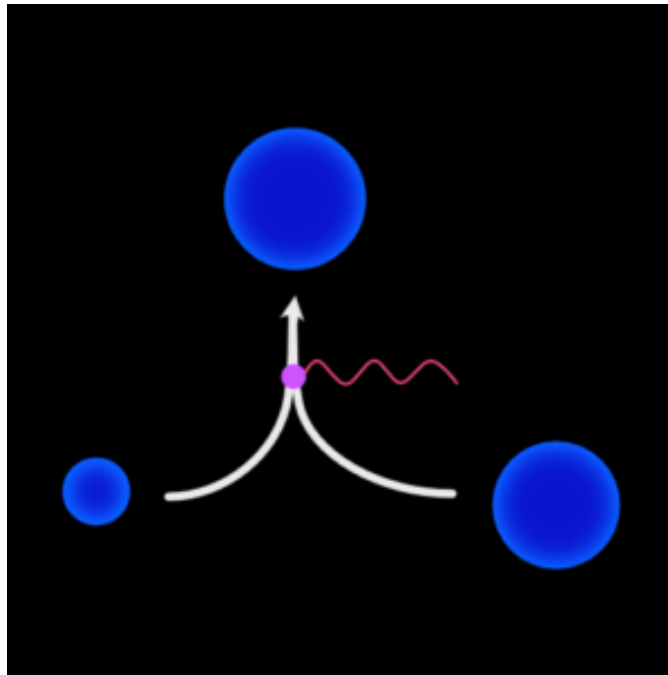
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[Step by step from Newton to Einstein](#)

On Earth and generally everywhere in our solar system, the gravitational field is, for cosmic terms, weak. Thanks to that circumstance, the gravitational field and as a consequence also the motion of the celestial bodies of the solar system can be described very well ...

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[Gravitational wave detectors find 56 potential cosmic collisions](#)

The epoch of gravitational wave astronomy has begun, concluded David Reitze, Executive Director of the LIGO gravitational wave detectors, according to the scientific journal Nature Reviews Physics in April 2020. While the first two observation runs (O1: September 2015 ...

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[third-generation gravitational-wave detector](#)

Third-generation gravitational-wave detectors are supposed to measure gravitational waves with higher sensitivity than current detectors such as Advanced LIGO or Advanced Virgo. Examples for such projects are the Einstein telescope and Cosmic Explorer. More information ...

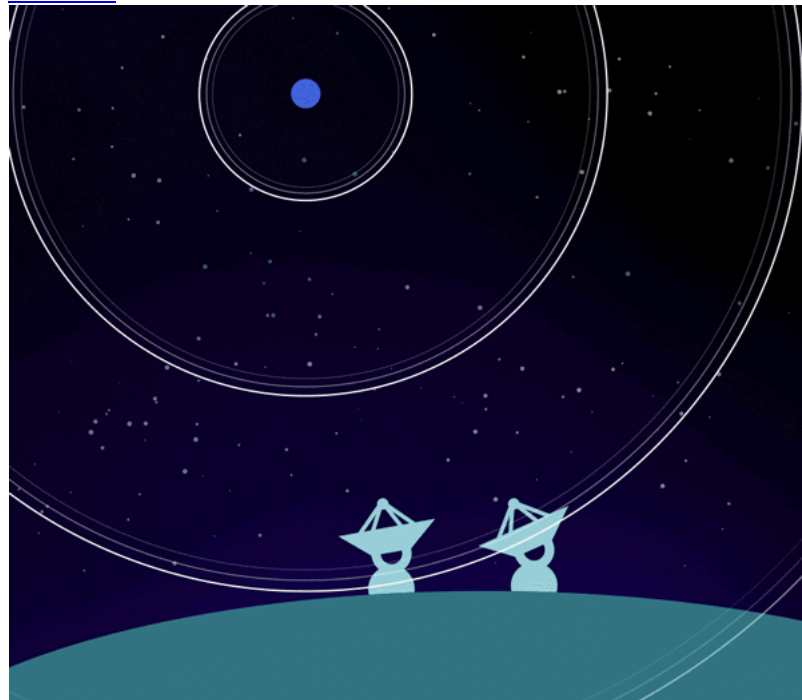
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Virgo

Virgo is a gravitational wave detector that was collaboratively built by French and Italian labs and was in operation until 2011. Then, the instrument was decommissioned and adapted to achieve 10 times higher sensitivity. Advanced Virgo started operation in 2017. ...

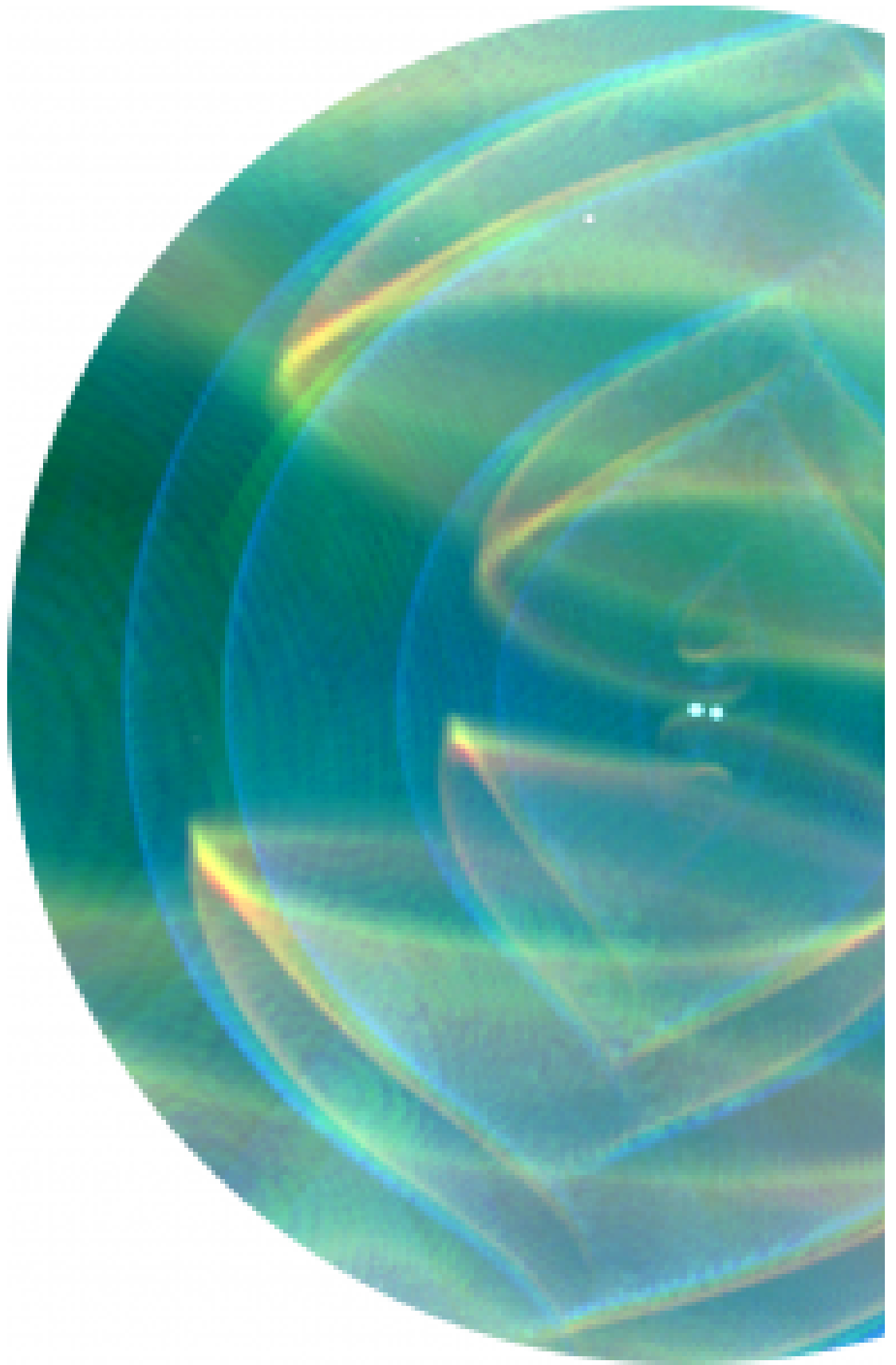
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Measuring light deflection with radio telescopes

The deflection of the path of light by mass is one of the classical predictions of general relativity. Probably its most famous successful test was the 1919 detection of the deflection of starlight by the Sun during a total solar eclipse, by comparing with similar ...

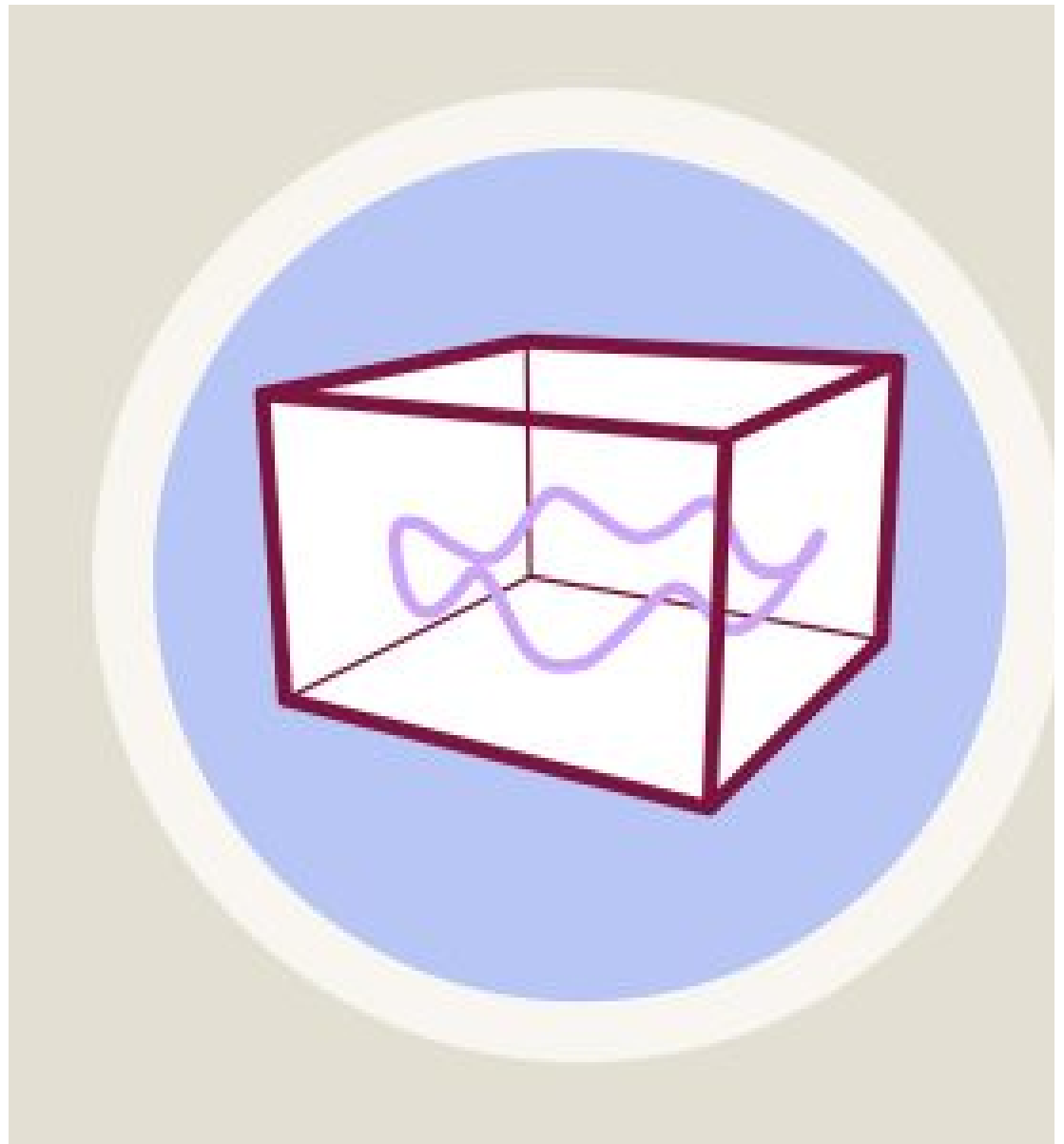
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[First measurement of gravitational waves of colliding neutron stars](#)

More than 100 years after Albert Einstein's formulation of the general theory of relativity and more than 30 years after the first discovery of a binary neutron star system, gravitational waves of colliding neutron stars have been detected for the first time. The ...

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Relativity and the Quantum / Elementary Tour part 5:
Superstrings and universal harmony.

If you go by the number of researchers involved in it, string theory is the most actively studied candidate for a theory of quantum gravity. String theory is descended from the ordinary models of particle physics, but with a crucial difference: Its basic constituents ...

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[Cosmology / Elementary Tour part 3: The early universe](#)

An expanding universe, with the distances between galaxy increasing all the time, must have been much more dense, and the galaxies much closer together, in the past. The details follow from Einstein's equations which connect the way expansions runs its course with the ...

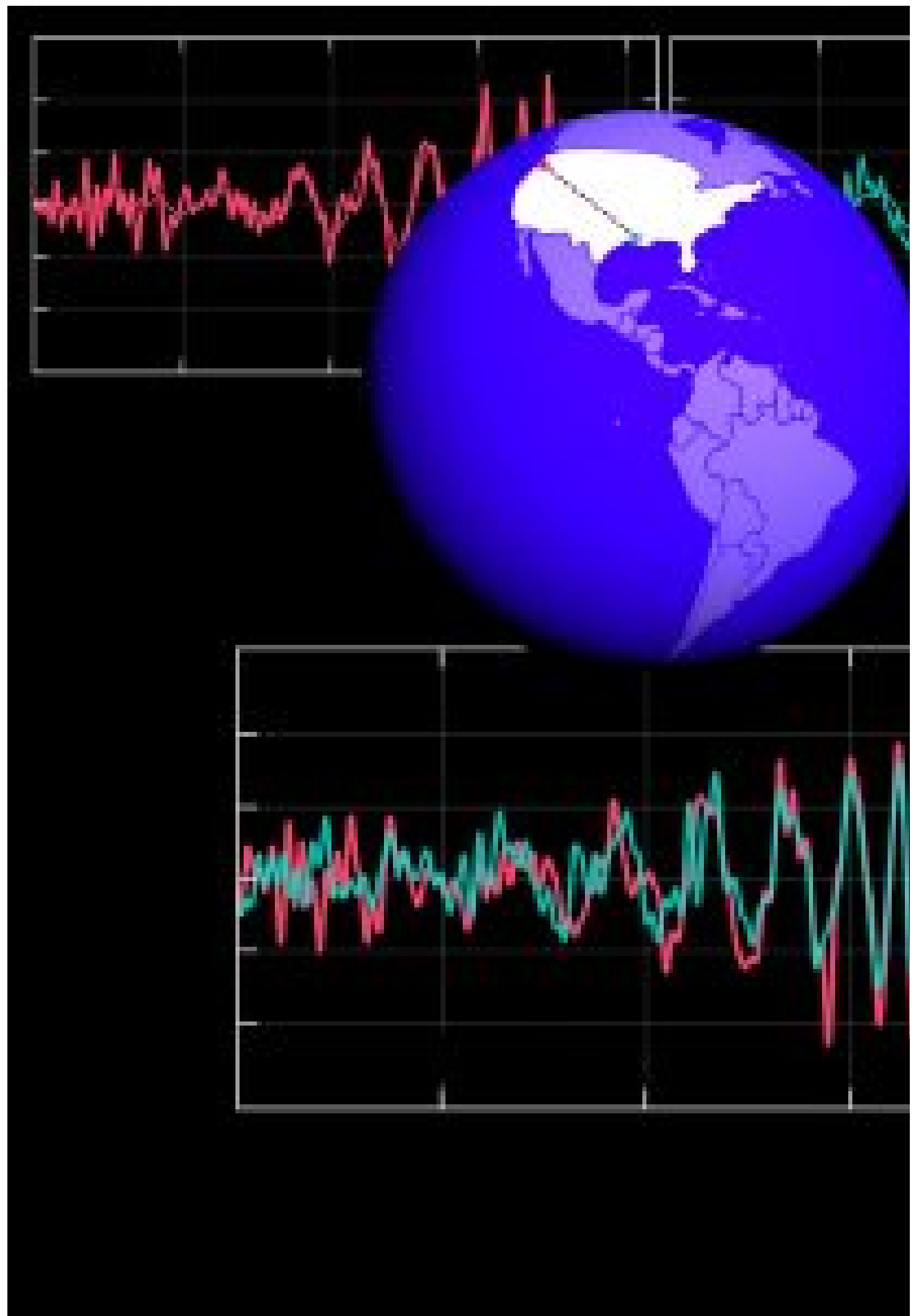
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[Cosmology / Elementary Tour part 2: Reddening galaxies](#)

Cosmic expansion has consequences for the light that reaches us from far-away galaxies - consequences that, in the 1920s, set astronomers on the track of an expanding universe in the first place. For an exact description of these effects, one would need a ...

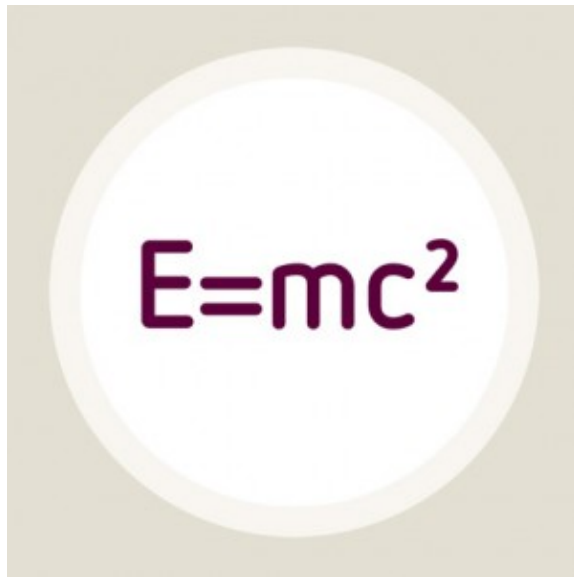
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Observation of Gravitational Waves from a Binary Black Hole Merger

One hundred years ago, Albert Einstein predicted in his general theory of relativity the existence of gravitational waves. These ripples in spacetime propagate at the speed of light and are generated by changes of the mass quadrupole moment of the source over time. ...

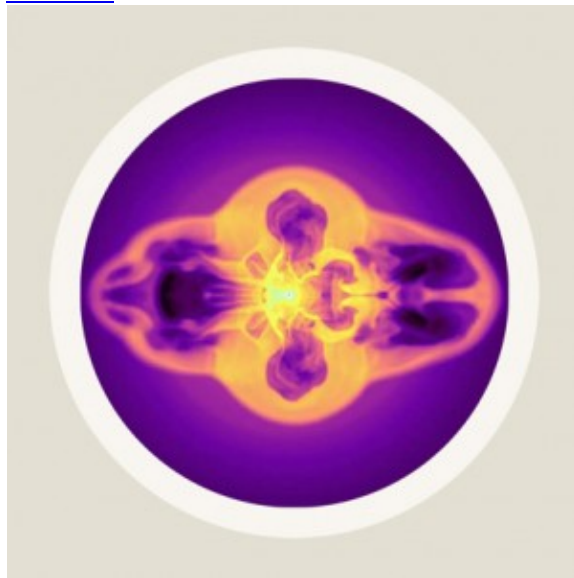
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[Special relativity / Elementary Tour part 6: \$E=mc^2\$](#)

Now that the new features of space and time are sufficiently explored, it's time to examine how this affects the way objects move, either freely or when influenced by forces. In other words: How does relativistic mechanics work? This question has led to the prediction ...

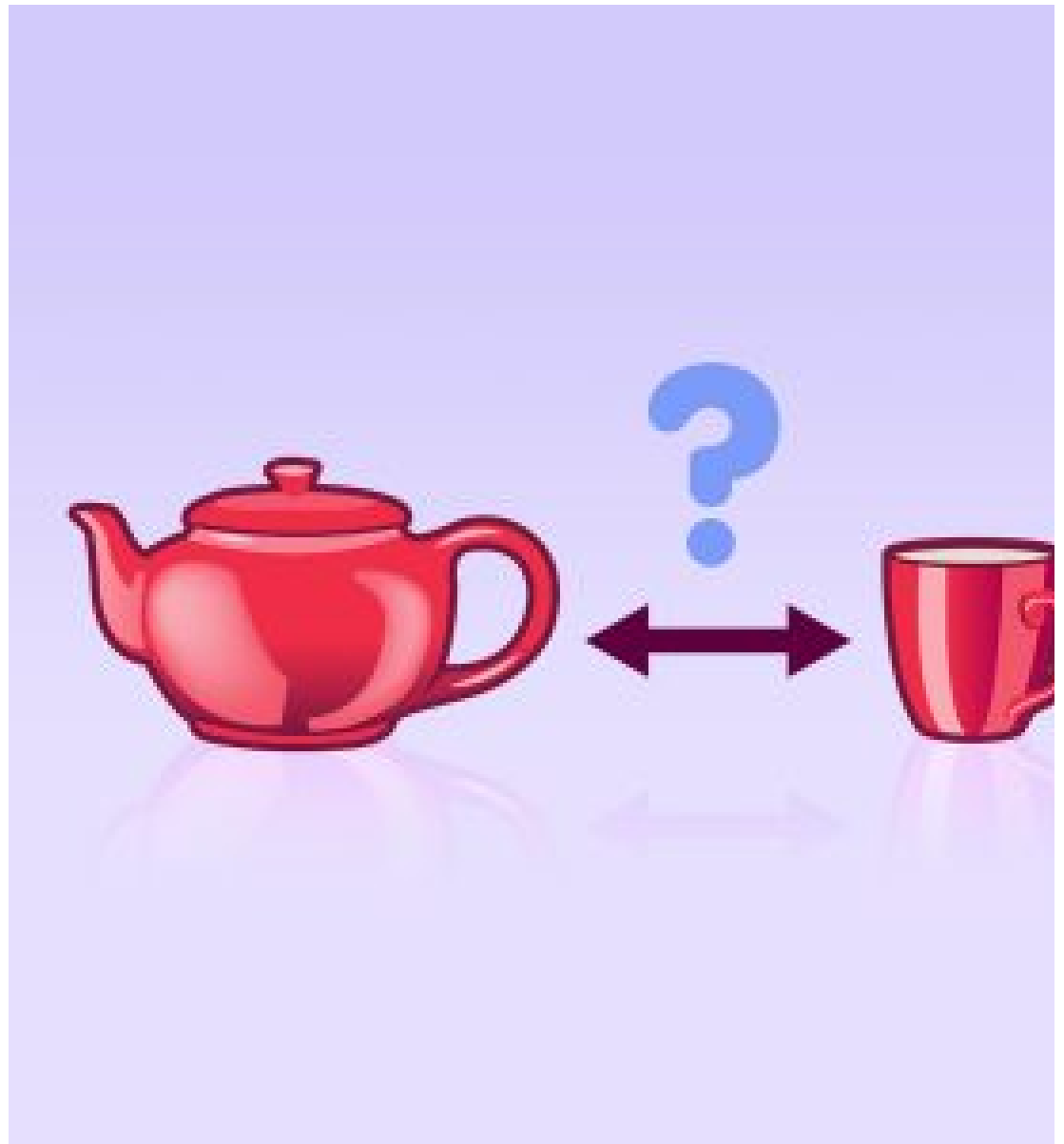
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[Gravitational waves / Elementary tour part 3: Gravitational wave astronomy](#)

Every single atom can radiate and absorb electromagnetic waves. The light reaching us from astronomical sources is a hodgepodge of little snippets of radiation, each snippet produced by one of the object's myriads of atoms. That's good news and bad news. The good ...

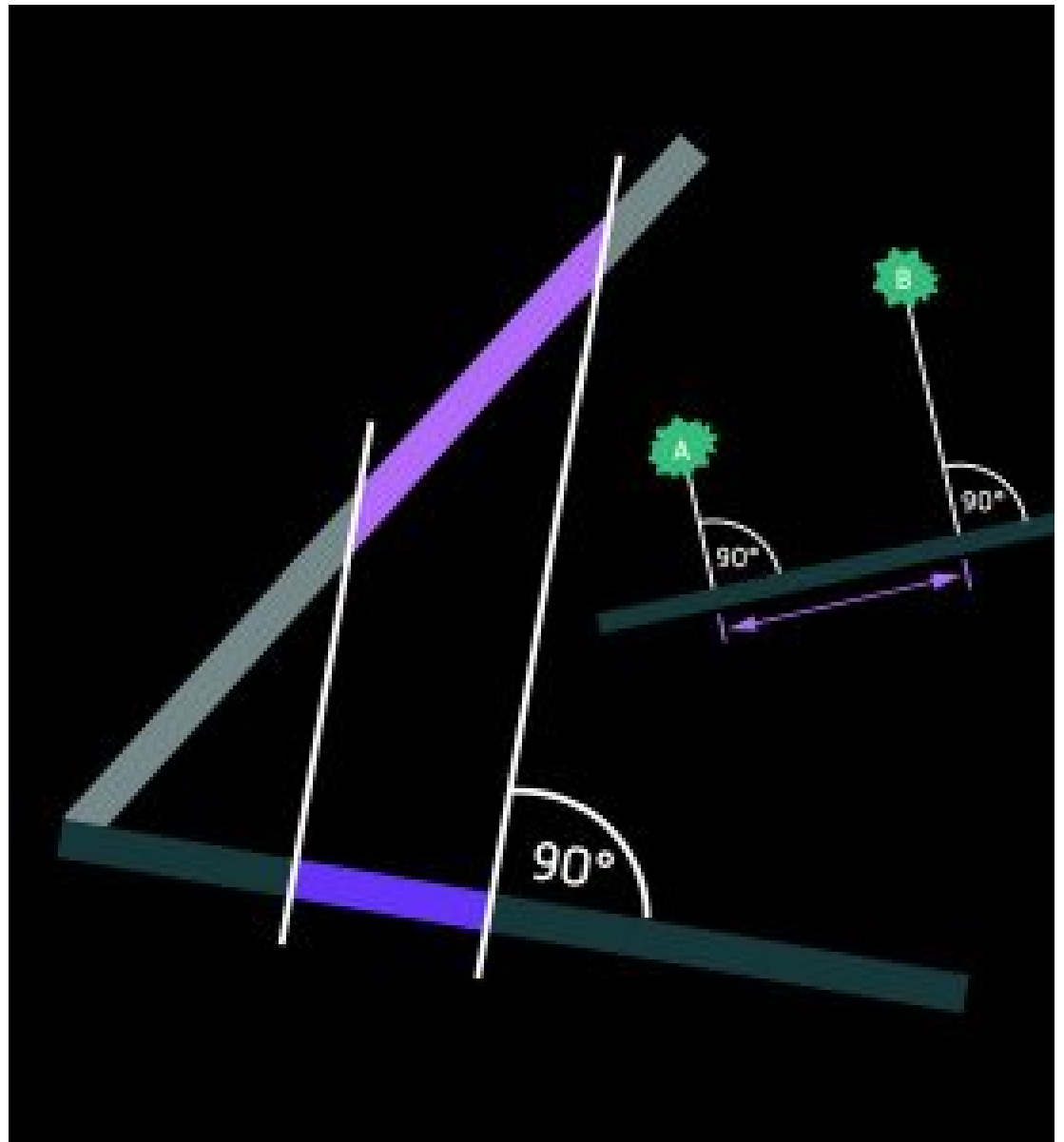
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[The dialectic of relativity](#)

When first encountered, some of the statements of special relativity sound downright paradoxical. However, in most cases the suspected paradox can be resolved within a framework of a "dialectic of relativity". (Here, the term "dialectic", which is borrowed from ...

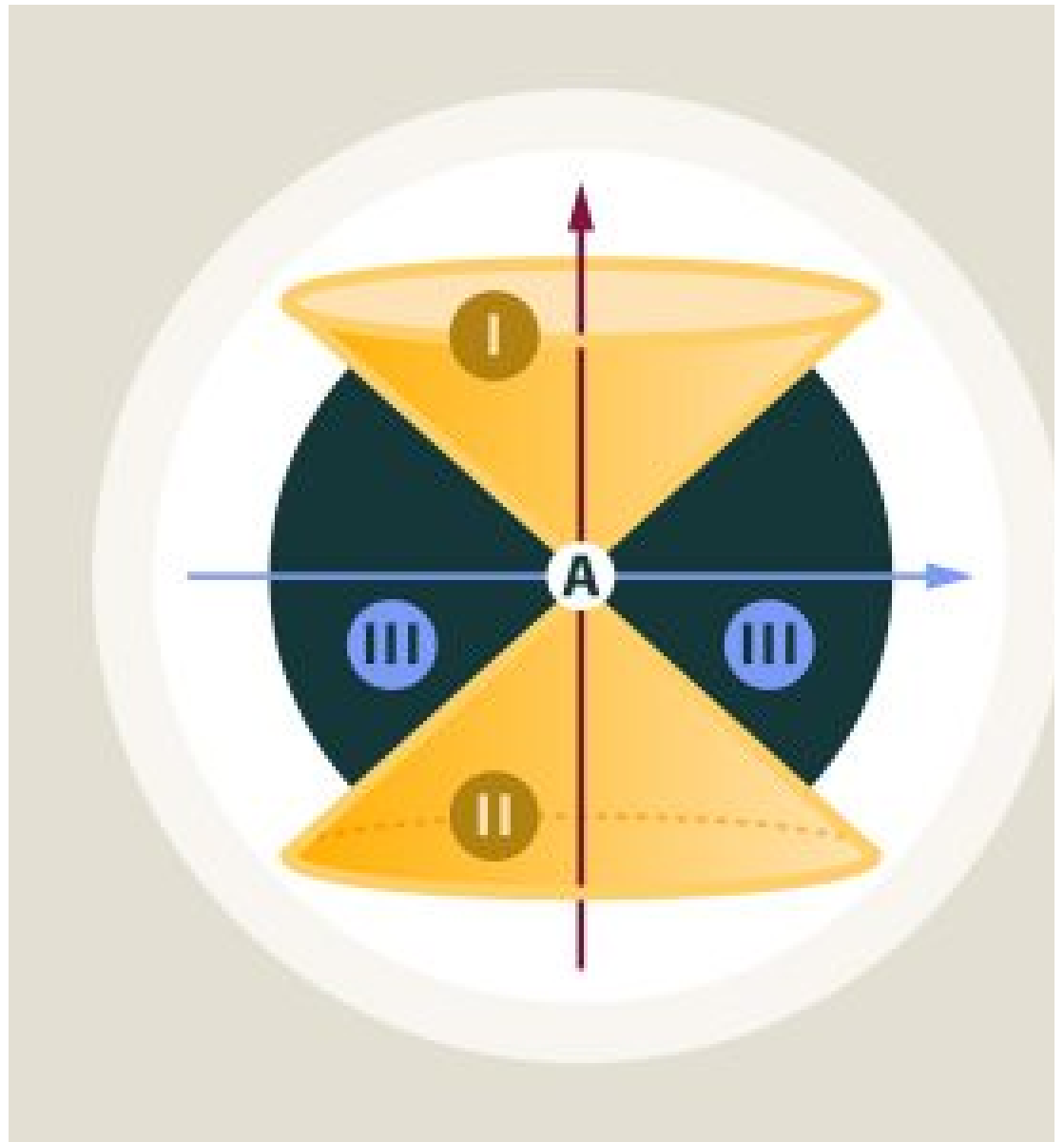
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[Time dilation on the road](#)

For a number of relativistic phenomena, there are helpful geometric analogies. They transplant space-time-phenomena to the more familiar arena of space geometry. Most people might not be familiar with space-time-lines and -projections, but they do know, for instance, ...

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[Special relativity / Elementary Tour part 5: Spacetime](#)

"All the world's a stage." - that's how we're used to viewing space: As a stage on which objects are located and where the dramas of their movements and evolution take place. In special relativity, as was mentioned briefly, simultaneity is relative, and so are time and ...

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[Heat that meets the eye](#)

Physical bodies (or, speaking more generally, physical systems) that are brought into close contact, close enough for them to exchange energy, evolve towards a state known as . In this state, all the systems involved have the same temperature - which is equivalent to ...

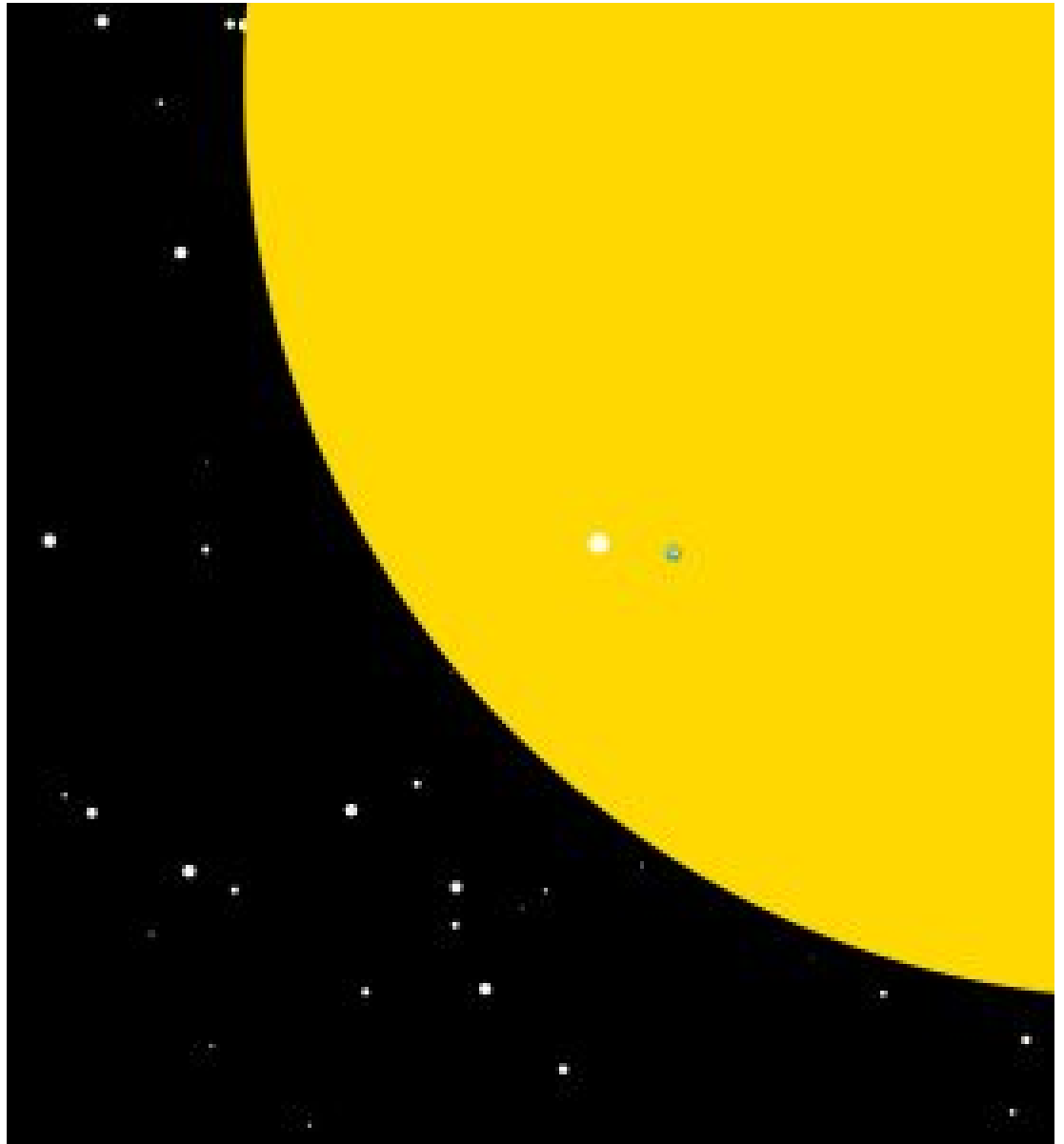
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[From soap bubbles to Einstein](#)

But children are not the only ones who are interested in soap bubbles. These fragile entities have captured the imagination of mathematicians, as well. From their physicist colleagues, mathematicians know which forces act on a soap bubble. Namely, there are ...

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[Gravitational redshift and White Dwarf stars](#)

One of the three classical tests for general relativity is the gravitational redshift of light or other forms of electromagnetic radiation. However, in contrast to the other two tests – the gravitational deflection of light and the relativistic perihelion shift –, ...

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What is Einstein Online?

Einstein Online is a web portal with comprehensible information on Einstein's theories of relativity and their most exciting applications from the smallest particles to cosmology.

Einstein Online is provided by the

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[Glossary](#)

More than 400 entries from "absolute zero" to "XMM Newton" - whenever you see [this type of link](#) on an Einstein Online page, it'll take you to an entry in our relativistic dictionary.

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Our recommendations for [books](#) and [websites](#) on relativity and its history.

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